

Q. Why do we consider the priority min-heap as the set data structure to find the shortest path while using Dijkstra's algorithm?

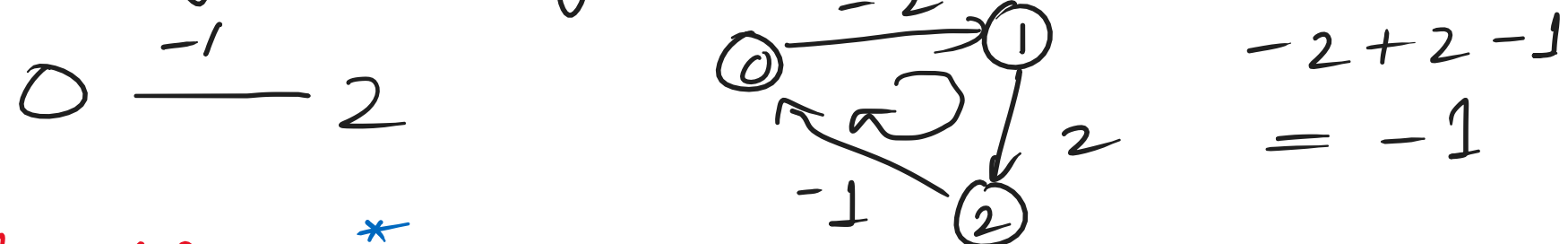
Ans: $\rightarrow$ We do so because if we use the queue data structure, we are forced to consider all the distances, rather for the pq $\rightarrow$ min-heap, Σ the set, we always consider the lowest / minimum distance for the traversal.

* Avoiding Unnecessary Iterations

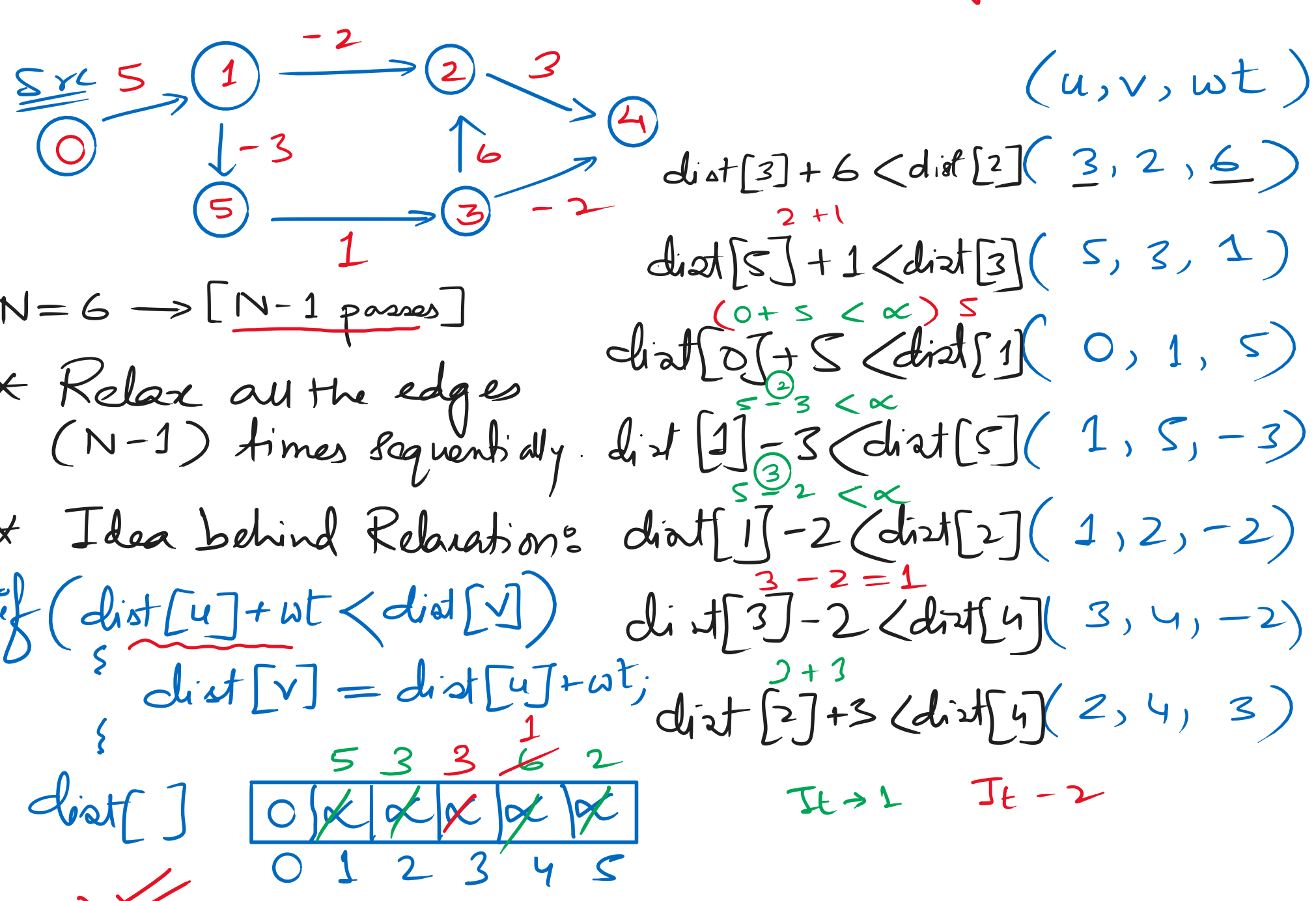
Bellman Ford Algorithm $\Rightarrow$ Shortest Path

* Negative Edge Weights are considered

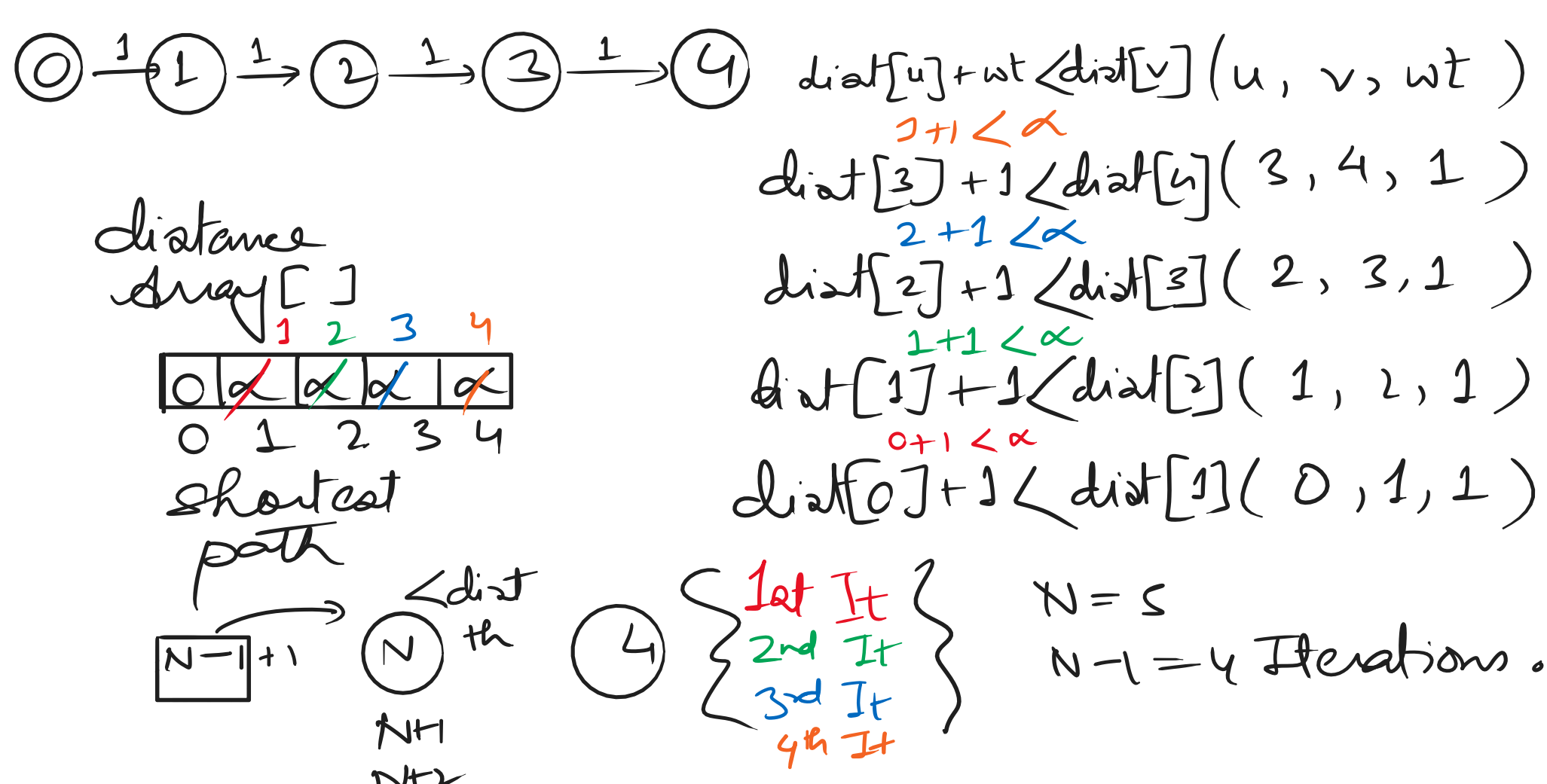
* Negative cycles are considered



graph $\rightarrow$ Shortest Distance $\uparrow \downarrow \uparrow \downarrow$
Progress $\rightarrow$ Monthly $\uparrow \downarrow \uparrow \downarrow$



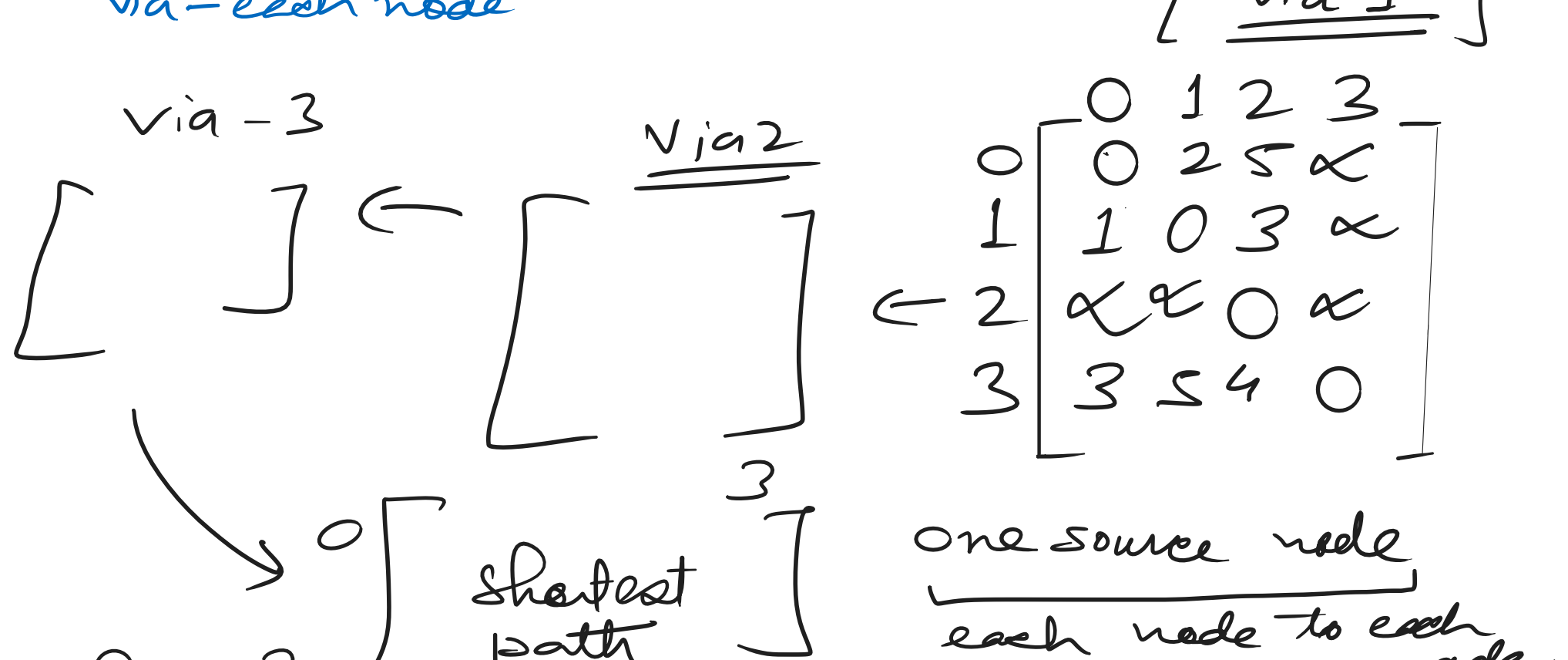
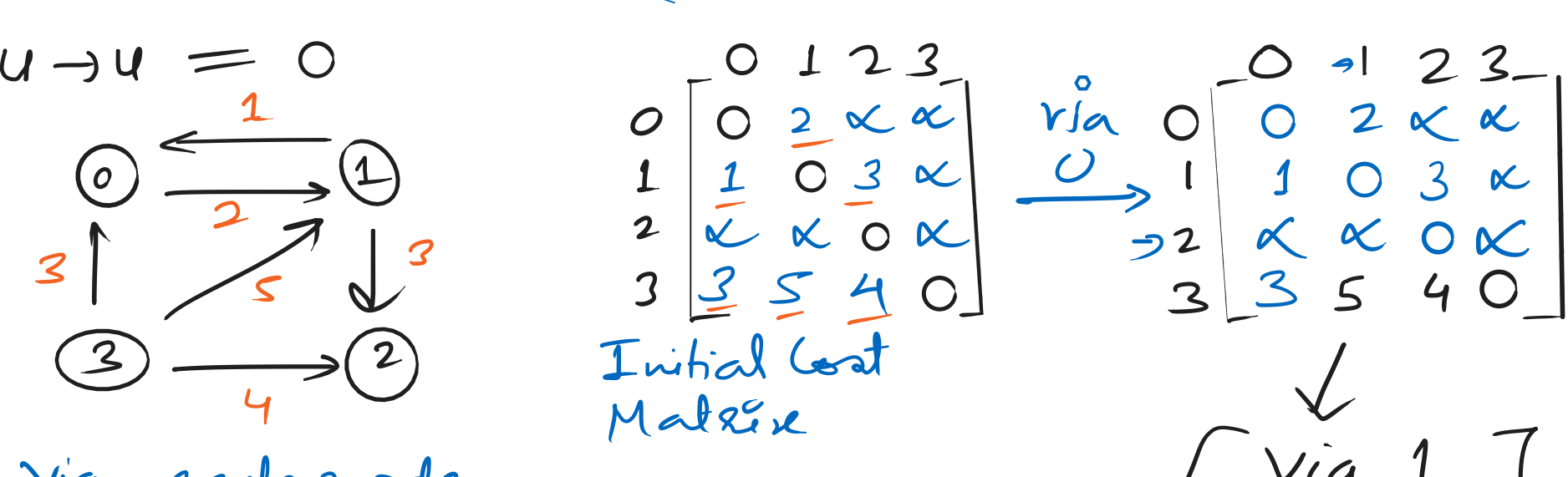
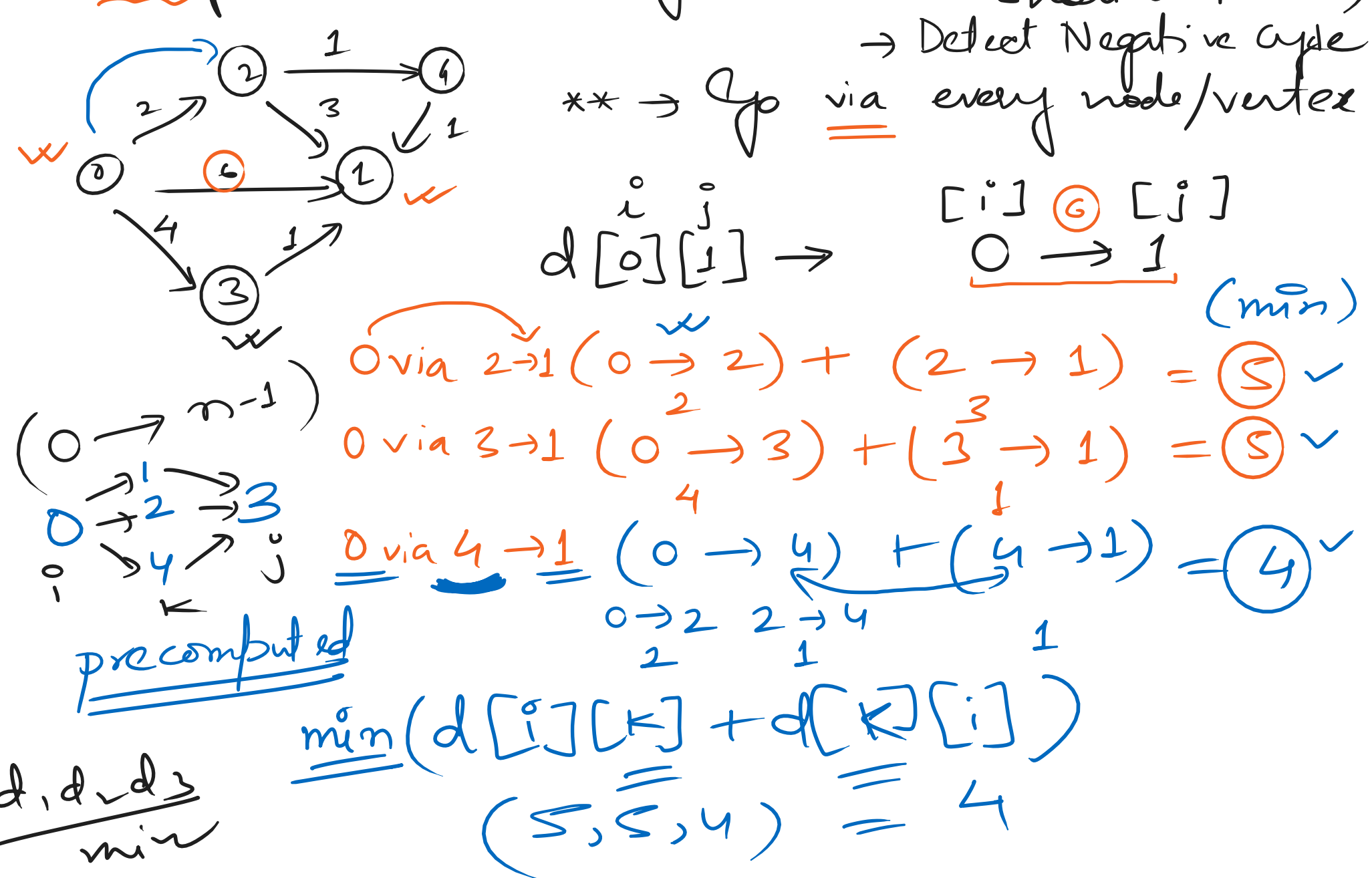
How does Bellman Ford complete in $N-1$ iterations? Also, how do we detect negative cycles in this algo?



On the N^{th} iteration, if the relaxation is still giving us reduced distances, it means that there is a negative cycle in the graph.

Because, the relaxation of all the nodes should complete in the $(N-1)^{th}$ iteration.

Floyd Warshall Algorithm $\Rightarrow$ (Multi-source Shortest Path)



[0 - n all the nodes] = via
for (via = 0; via < n; via++)
{
 for (i = 0; i < n; i++)
 for (j = 0; j < n; j++)
 {
 cost[i][j] = min(cost[i][j],
 cost[i][via] + cost[via][j])
 }
}

How to detect negative cycle?

